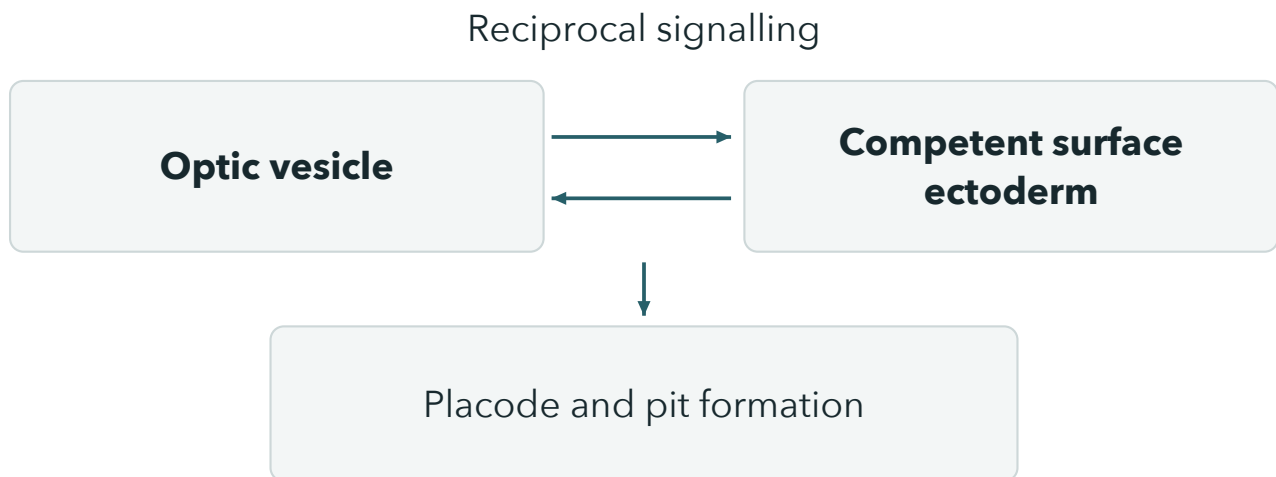


Developmental signalling and tissue interaction

Part 1 | Induction requires interaction

Competent ectoderm and optic-vesicle signals contribute different parts of lens induction.



Later: closed vesicle separates from surface

Reciprocal developmental interaction is shown separately from later vesicle separation.

A staged experiment

Preventing optic-vesicle contact can impair placode formation despite preserved neural tissue. A later ectodermal intervention need not produce the same phenotype. Compare tissue competence, signalling and timing before calling a defect a failure of induction.

From placode to separate vesicle

Formation and separation are successive events, not interchangeable descriptions.

Morphological sequence

- Surface ectoderm thickens into placode
- ↓
- Placode invaginates into lens pit
- ↓
- Lens vesicle closes
- ↓
- Vesicle separates from surface ectoderm

Observed failure

What is present?	First suspect process
No placode	Induction/competence
Placode without invagination	Morphogenesis
Closed vesicle with persistent bridge	Separation

Read the observation literally

A closed lens vesicle establishes prior induction and invagination. A persistent epithelial bridge points to failed separation, not merely an abnormal tissue origin.

Anterior-segment dysgenesis

A phenotype can reflect disrupted tissue interaction rather than one missing derivative.

Affected system

Feature	Developmental contribution	Potential disturbance
Corneal stroma/endothelium	Periocular mesenchyme	Migration or differentiation
Lens and corneal interface	Tissue separation	Persistent adhesion
Angle architecture	Coordinated anterior development	Abnormal outflow structures

Mechanistic caution

Peters anomaly has more than one described developmental mechanism.

A lens-cornea adhesion does not identify a single mutation.

The phenotype should guide, not replace, molecular interpretation.

Compare preserved structures

A formed lens and differentiated retina help locate the stage of failure. They do not establish that every signalling interaction between these tissues was normal.

Patterning genes are not isolated labels

Transcription factors regulate networks and can affect several ocular structures.

Evidence level

Observation	Permitted inference
One developmental phenotype	Network disturbance possible
Pathogenic variant plus phenotype	Genotype-phenotype support
Negative limited panel	A genetic cause remains possible

Think beyond one structure

PAX6-associated disease can involve more than the visible iris.

FOXC1 and PITX2 participate in anterior-segment development.

Variable expressivity can produce different severity within a family.

Useful boundary

Do not interpret a normal-appearing iris as proof that an entire developmental pathway is intact. Equally, one anterior-segment finding is not sufficient to assign a molecular diagnosis.

Matrix participates in morphogenesis

Developmental shape depends on mechanics as well as gene expression.

An experimental hypothesis

- Tissues establish a contact region
↓
- Extracellular matrix constrains local expansion
↓
- Cells change shape and packing
↓
- Placode and pit geometry emerge

What a rescue experiment adds

Restoring matrix support can test a mechanical contribution.

Restoring a transcription factor tests a different level of the network.

Similar final phenotypes can arise through different upstream failures.

Avoid a single-cause story

Placode formation involves signalling, competence and tissue mechanics. A manipulation that alters one variable is strongest when the other variables are measured, rather than assumed unchanged.